longops

Long operations and detailed calculations.
Opérations posées et calculs détaillés.

v0.1.1 Akilon <https://github.com/Akilon27> MIT

About

As a mathematics teacher, I sometimes need to show how to carry out arithmetic “by hand”. I am now making available the functions that I originally created for my own use.

I previously used the LaTeX package **xlop**, and since Typst is much more convenient to use, I decided to create my own equivalent. This package is **not** a direct translation of **xlop**: it includes some features that **xlop** does not, while omitting others, according to my own needs.

I occasionally used AI tools (mainly Gemini for the **Order of Operations** module, and also to add features to the other functions that I had initially implemented for just one of them).

Acknowledgments : I wanted to thank ObaulG, the part “durations” is mostly is work.

Functions for handwritten arithmetic layouts ...

- add-en-ligne()
- addition()
- division()
- egalite-decimale()
- egalite-euclidienne()
- multiplication()
- racine()
- sous-en-ligne()
- soustraction()

add-en-ligne

Compute the sum of all the supplied numbers inline.

```
$#add-en-ligne(198.7,120,35)$
```

$198.7 + 120 + 35 = 353.7$

Parameters

```
add-en-ligne(..ops: arguments ) -> content
```

addition

Lay out the column addition of 2 or more numbers with the possibility of hiding the result with `hide-result`, or a `liste` of some of the digits and then to show the `solution`, a few colors can be set

Examples :

```
#addition(12.5, 3.75, .8, 14.2)
```

```
   1 2
   1 2 , 5 0
+    3 , 7 5
+    0 , 8 0
+  1 4 , 2 0
-----------
   3 1 , 2 5
```

```
#addition(12.5, 3.75, .8, 14.2,liste:
(2,10,19))
```

```
   1 ▢ , 5 0
+    3 , ▢ 5
+    0 , 8 0
+  ▢ 4 , 2 0
-----------
   3 1 , 2 5
```

Parameters

```
addition(
  show-carry: boolean,
  couleur: color,
  size: length,
  fr: boolean,
  hide-result: boolean,
  couleur-cadre: color,
  couleur-solution: color,
  type-mask: str,
  liste: array,
  solution: boolean,
  signe: boolean content,
  space: length,
  texte: length,
  ..args: array arguments
) -> content
```

show-carry `boolean`

show or hide the carries

Default: `true`

couleur `color`

color of the carries

Default: `red`

size `length`

Size of the columns

Default: `1em`

fr `boolean`

french decimal separaro "," or not

Default: `true`

hide-result `boolean`

Hide the carry row and the result

Default: `false`

couleur-cadre `color`

Color of the mask of hidden digits

Default: `blue.mix(gray)`

couleur-solution `color`

Color of the hidden digits if solution = true

Default: `red`

type-mask `str`

How the digits are hidden : “rect” for a round rectangle or “” for dots

Default: `"rect"`

liste `array`

list of the digits to hide

Default: `()`

solution `boolean`

show the carries and the hidden digits

Default: `false`

signe `boolean` or `content`

Sign of equality, false or a symbol $ = $

Default: `false`

space `length`

Space if more than one calculation on a line, default : 1fr

Default: `1fr`

texte `length`

size of the text : 1em

Default: `1em`

division

Divide decimal number `a` by `b`.

With `extradigits`, computes additional digits after the decimal point beyond those already present in `a` (when `cycle:false`). These extra digits are displayed in gray (`mode:"g"` by default), white (`mode:"0"`), as dots (`mode:"p"` or `"gp"`), or omitted (`mode:""`). Use `s:true/false` to show or hide the subtraction steps. You can highlight the repeating cycle in the quotient using `cycle:true` (underlined with `cycle-color`), display it in parentheses with `cycle:""`, or display it in color with `cycle:red`. Fill-in-the-blank divisions are also supported with `hide-result`, `liste`, and `solution` (using the `type-mask`, `couleur-cadre`, and `couleur-solution` parameters).

Examples:

```
#division(12.6, 3.75)
```

```
 1260|375
-1125|3
  135|
```

```
#division(121.5, 7, cycle:true)
```

```
121,5000000|7
-7         |17,3571428
 51        |
-49        |
  25       |
 -21       |
   40      |
  -35      |
    50     |
   -49     |
     10    |
     -7    |
      30   |
     -28   |
       20  |
      -14  |
        60 |
       -56 |
         4 |
```

```
#division(121.5, 7,
cycle:true,fr:false,dx:.5em,dy:-.9em)
```

```
   17.3571428
7)121.5000000
 -7
  51
 -49
   25
  -21
    40
   -35
     50
    -49
      10
      -7
       30
      -28
        20
       -14
         60
        -56
          4
```

Parameters

```
division(
  a: int float,
  b: int float,
  mode: str,
  extradigits: int,
  s: boolean,
  cycle: boolean str color,
  cycle-color: color,
  fr: boolean,
  size: length,
  dx,
  dy,
  hide-result: boolean,
  liste: array,
  solution: boolean,
  type-mask: str,
  couleur-cadre: color,
  couleur-solution: color,
  space: length
) -> content
```

mode `str`

Display mode for zeros added to the dividend: “g” (gray), “0” (black), “p” (small dot), “gp” (large dot), “” (empty)

Default: `"g"`

extradigits `int`

Number of additional quotient digits to compute

Default: `0`

s `boolean`

Show or hide the subtraction steps

Default: `true`

cycle `boolean` or `str` or `color`

Detect and display the repeating cycle automatically

Default: `false`

cycle-color `color`

Color of the cycle or underline

Default: `luma(50%)`

fr `boolean`

If fr:true, the decimal separator is “,” otherwise “.”

Default: `true`

size `length`

Column width

Default: `.7em`

dx

dx for the) in mode fr:false

Default: `.6em`

dy

dy for the) in mode fr:false

Default: `-.82em`

hide-result `boolean`

Hide the working steps and quotient (exercise mode)

Default: `false`

liste `array`

List of cell indices to hide (dividend, divisor, steps, quotient)

Default: `()`

solution `boolean`

Show or hide the solution inside the masks

Default: `false`

type-mask `str`

Mask type (“rect” or “line”)

Default: `"rect"`

couleur-cadre `color`

Color of the mask border

Default: `blue.mix(gray)`

couleur-solution `color`

Color of the solution text

Default: `red`

space `length`

Spacing when several calculations are on the same line

Default: `1fr`

egalite-decimale

Write the equality corresponding to a decimal division.

```
#egalite-decimale(126, 25)
```

$126 \div 25 = 5.04$

Parameters

```
egalite-decimale(
  a: int float,
  b: int float,
  s: str
) -> content
```

egalite-euclidienne

Write the equality corresponding to a Euclidean division.

```
#egalite-euclidienne(126, 27)
```

$126 = 4 \times 27 + 18$

Parameters

```
egalite-euclidienne(
  a: int float,
  b: int float
) -> content
```

multiplication

Lay out the multiplication of 2 integer or decimal numbers with the possibility of hiding intermediate rows using `hide-result`, hiding a `liste` of selected digits, then displaying the `solution`, with various configurable colors.

Examples:

```
#multiplication(12.5, 3.75)
```

```
       12,5
×     3,7 5
      6 2 5
+   8 7 5 0
+ 3 7 5 0 0
= 46,8 7 5
```

```
#multiplication(1215, 142,liste:
(3,12,19,36),type-mask: "line",)
```

```
        … 2 1 5
×         1 … 2
      2 4 … 0
+   4 8 6 0 0
+ 1 2 1 5 0 0
  1 7 2 5 3 0
```

Parameters

```
multiplication(
  a: int float,
  b: int float,
  mode: str,
  sym: boolean str,
  fr: boolean str,
  size: length,
  hide-result: boolean,
  liste: array,
  solution: boolean,
  type-mask: str,
  couleur-cadre: color,
  couleur-solution: color,
  space: length
) -> content
```

mode `str`

Display mode for shifted rows: “g” (default) for gray zeros, “0” for normal zeros, “p” for small dots, “gp” for larger dots, “” or anything else for nothing

Default: `"g"`

sym `boolean` or `str`

Show the addition “+” signs and the final “=” if true; only “+” if “+”, only “=” if “=”, otherwise nothing

Default: `true`

fr `boolean` or `str`

If fr:true, the decimal separator is “,” otherwise “.”

Default: `true`

size `length`

Column width

Default: `1em`

hide-result `boolean`

Hide intermediate rows and the final result for exercises

Default: `false`

liste `array`

List of cell indices to hide (fill-in-the-blank multiplication)

Default: `()`

solution `boolean`

Show or hide the solution inside the masks

Default: `false`

type-mask `str`

Mask type (“rect” or “line”)

Default: `"rect"`

couleur-cadre `color`

Color of the mask border

Default: `blue.mix(gray)`

couleur-solution `color`

Color of the solution text

Default: `red`

space `length`

Spacing when several calculations are on the same line

Default: `1fr`

racine

Compute the square root of `a`, with `extradigits` additional decimal places, optionally showing digit pairs using `groupes` in `couleur-groupes`, displaying the construction step by step with `step`, hiding subtraction steps with `s`, and hiding the result or a `liste` of digits.

Examples:

```
#racine(24368,size: 1em,liste: (2,27))
```

```
 2 4 ▢ 6 8 | 1 ▢ 6
−1          | 1² = 1 < 2
 1 4 3      | 25 × 5 = 125
−1 2 5      | 306 × 6 = 1836
   1 8 6 8  |
  −1 8 3 6  |
       3 2  |
```

```
#grid(columns: 3*(1fr,), row-gutter: 1em,
 ..for i in range(1,11) {([
   ==== Étape #i
   #racine(24368, extradigits: 1,groupes: true, step: i)
 ],)+if i == 1 {([],[],)}}
 )
```

Étape 1

$$\begin{array}{r|l}
24368 & 1 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
1 &
\end{array}$$

Étape 2

$$\begin{array}{r|l}
24368 & 1 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 2{\color{blue}c} \times {\color{blue}c} \leqslant 143
\end{array}$$

Étape 3

$$\begin{array}{r|l}
24368 & 1 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 25 \times {\color{blue}5} = 125
\end{array}$$

Étape 4

$$\begin{array}{r|l}
24368 & 15 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 25 \times {\color{blue}5} = 125 \\
-\underline{125} & \\
18 &
\end{array}$$

Étape 5

$$\begin{array}{r|l}
24368 & 15 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 25 \times {\color{blue}5} = 125 \\
-\underline{125} & 30{\color{blue}c} \times {\color{blue}c} \leqslant 1868 \\
1868 &
\end{array}$$

Étape 6

$$\begin{array}{r|l}
24368 & 15 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 25 \times {\color{blue}5} = 125 \\
-\underline{125} & 306 \times {\color{blue}6} = 1836 \\
1868 &
\end{array}$$

Étape 7

$$\begin{array}{r|l}
24368 & 156 \\ \cline{2-2}
-\underline{1} & 1^2 = 1 < 2 \\
143 & 25 \times {\color{blue}5} = 125 \\
-\underline{125} & 306 \times {\color{blue}6} = 1836 \\
1868 & \\
-\underline{1836} & \\
32 &
\end{array}$$

Étape 8

$$\begin{array}{r|l}
24368{,}00 & 156 \\ \cline{2-2}
-\underline{1}\phantom{4368{,}00} & 1^2 = 1 < 2 \\
143\phantom{68{,}00} & 25 \times {\color{blue}5} = 125 \\
-\underline{125}\phantom{68{,}00} & 306 \times {\color{blue}6} = 1836 \\
1868\phantom{{,}00} & 312{\color{blue}c} \times {\color{blue}c} \leqslant 3200 \\
-\underline{1836}\phantom{{,}00} & \\
3200 &
\end{array}$$

Étape 9

$$\begin{array}{r|l}
24368{,}00 & 156 \\ \cline{2-2}
-\underline{1}\phantom{4368{,}00} & 1^2 = 1 < 2 \\
143\phantom{68{,}00} & 25 \times {\color{blue}5} = 125 \\
-\underline{125}\phantom{68{,}00} & 306 \times {\color{blue}6} = 1836 \\
1868\phantom{{,}00} & 3121 \times {\color{blue}1} = 3121 \\
-\underline{1836}\phantom{{,}00} & \\
3200 &
\end{array}$$

Étape 10

$$\begin{array}{r|l}
24368{,}00 & 156{,}1 \\ \cline{2-2}
-\underline{1}\phantom{4368{,}00} & 1^2 = 1 < 2 \\
143\phantom{68{,}00} & 25 \times {\color{blue}5} = 125 \\
-\underline{125}\phantom{68{,}00} & 306 \times {\color{blue}6} = 1836 \\
1868\phantom{{,}00} & 3121 \times {\color{blue}1} = 3121 \\
-\underline{1836}\phantom{{,}00} & \\
3200 & \\
-\underline{3121} & \\
79 &
\end{array}$$

Parameters

```
racine(
  a: int float,
  extradigits: int,
  groupes: boolean,
  couleur-groupes: color,
  step: none int,
  couleur-step: color,
  s: boolean,
  fr: boolean,
  size: length,
  hide-result: boolean,
  liste: array,
  solution: boolean,
  type: str,
  couleur-cadre: color,
  couleur-solution: color,
  space: length,
  ..args: arguments
) -> content
```

extradigits `int`

Number of additional “00” digit pairs to compute after the decimal point

Default: `0`

groupes `boolean`

Show brackets above each pair of digits

Default: `false`

couleur-groupes `color`

Color of the brackets

Default: `blue.darken(20%)`

step `none` or `int`

Step to display in the step-by-step construction

Default: `none`

couleur-step `color`

Color of the step indicator

Default: `blue.darken(20%)`

s `boolean`

Show or hide the subtraction steps below the dividend

Default: `true`

fr `boolean`

If fr:true, the decimal separator is "," otherwise "."

Default: `true`

size `length`

Column width

Default: `0.65em`

hide-result `boolean`

Hide the construction steps, operations, and result (exercise mode)

Default: `false`

liste `array`

List of cell indices to hide (fill-in-the-blank exercise)

Default: `()`

solution `boolean`

Show or hide the solution inside the masks

Default: `false`

type `str`

Mask type ("rect" or "line")

Default: `"rect"`

couleur-cadre `color`

Color of the mask border

Default: `blue.mix(gray)`

couleur-solution `color`

Color of the solution text

Default: `red`

space `length`

Right spacing

Default: `1fr`

sous-en-ligne

Compute the difference of all the supplied numbers inline.

```
$#sous-en-ligne(198.7,107.3,-35)$
```

$198.7 - 107.3 - (-35) = 126.4$

Parameters

```
sous-en-ligne(
  ..ops: arguments,
  digits: auto int
) -> content
```

soustraction

Lay out the subtraction of 2 or more numbers with the possibility of hiding the result row with `hide-result`, hiding a `liste` of selected digits, then displaying the `solution`, with various configurable colors.

Examples:

```
#soustraction(12.5, 3.75)
```

```
   1 12 , 15 10
−     3 , 7  5
  +1 +1    +1
  ------------
      8 , 7  5
```

```
#soustraction(121.5, 3.75, .8,
14.2,liste:(2,10,19))
```

```
  1 □ 1 , 5 0
−     □ , 7 5
−     0 , □ 0
−   1 4 , 2 0
  -----------
  1 0 2 , 7 5
```

Parameters

```
soustraction(
  show-borrow: boolean,
  barre: boolean,
  fr: boolean,
  couleur: color,
  size: length,
  zeros: boolean,
  hide-result: boolean,
  couleur-cadre: color,
  couleur-solution: color,
  type-mask: str,
  liste: array,
  solution: boolean,
  signe: boolean content,
  texte: length,
  space: length,
  ..args: array arguments
) -> content
```

show-borrow `boolean`

Show or hide the borrows

Default: `true`

barre `boolean`

If fr:false, show or hide the crossed-out original digit

Default: `false`

fr `boolean`

French notation or English borrowing method

Default: `true`

couleur `color`

Color of the borrows

Default: `red`

size `length`

Column width

Default: `1em`

zeros `boolean`

Add trailing zeros when needed

Default: `true`

hide-result `boolean`

Hide the borrows and the result

Default: `false`

couleur-cadre `color`

Color of the border around hidden digits

Default: `blue.mix(gray)`

couleur-solution `color`

Color of hidden digits when solution = true

Default: `red`

type-mask `str`

How hidden digits are displayed: “rect” for a rounded rectangle or ...

Default: `"rect"`

liste `array`

List of digits to hide in a fill-in-the-blank subtraction

Default: `()`

solution `boolean`

Show the borrows and the result in hide-result mode (for answer keys)

Default: `false`

signe `boolean` or `content`

Equality sign, false or a symbol $ = $

Default: `false`

texte `length`

Text size: 1em

Default: `1em`

space `length`

Spacing when several calculations are on the same line

Default: `1fr`

Additions and substractions of durations

- addition-durees()
- prepa-duree()
- soustraction-durees()

addition-durees

Sets up the addition of two or more durations.

You need to change units: ([j], [h], [min], [s]) with units: ([d], [h], [min], [s]).

Each duration is an array in one of the following forms:

- `(days, hours, minutes, seconds)`.
- `(hours, minutes, seconds)`.
- `(minutes, seconds)`.

Examples:

```
#addition-durees(
  (1, 2, 45, 50),
  (0, 1, 20, 30),
)
```

	j	h	min	s
		1	1	
	1	02	45	50
+	0	01	20	30
	1	04	06	20

```
#addition-durees(
  (
    (1, 2, 45, 50),
    (0, 1, 20, 30),
    (2, 23, 55, 45),
  ),
  hide-result: true,
)
```

	j	h	min	s
	1	2	2	
	1	02	45	50
+	0	01	20	30
+	2	23	55	45

Parameters

```
addition-durees(
  show-carry: boolean,
  couleur: color,
  carry-size: relative length,
  size: relative length,
  convertion: boolean,
  hide-empty: boolean,
  hide-result: boolean,
  couleur-cadre: color,
  couleur-solution: color,
  type-mask: str,
  liste: array,
  solution: boolean,
  signe: boolean content,
  texte: relative length,
  show-units: boolean,
  units: array,
  units-size: relative length,
  couleur-unites: color,
  zero-pad: boolean,
  align: boolean,
  verticales: boolean,
  ..args: array arguments
) -> content
```

show-carry `boolean`

Show carries.

Default: `true`

couleur `color`

Color of the carries.

Default: `red`

carry-size `relative length`

Size of the carries and the + and = symbols, max. 1em.

Default: `.8em`

size `relative length`

Width of the columns corresponding to the units.

Default: `2.5em`

convertion `boolean`

Convert the result instead of displaying carries.

Default: `false`

hide-empty `boolean`

Hide columns containing only zeros.

Default: `true`

hide-result `boolean`

Hide the result.

Default: `false`

couleur-cadre `color`

Border color of hidden values.

Default: `blue.mix(gray)`

couleur-solution `color`

Color of values in the solution.

Default: `red`

type-mask `str`

Mask type: “rect” or “line”.

Default: `"rect"`

liste `array`

Indices of the cells to hide.

Default: `()`

solution `boolean`

Show the solution for hidden cells.

Default: `false`

signe `boolean` or `content`

Sign placed before the result.

Default: `false`

texte `relative length`

Text size.

Default: `1em`

show-units `boolean`

Show unit names.

Default: `true`

units `array`

Names of the four units.

Default: `([d], [h], [min], [s])`

units-size `relative length`

Unit size, max. 1em.

Default: `.75em`

couleur-unites `color`

Color of the units.

Default: `gray`

zero-pad `boolean`

Display hours, minutes, and seconds using two digits.

Default: `true`

align `boolean`

Align the digits of the units.

Default: `true`

verticales `boolean`

Vertical dotted lines to separate the columns.

Default: `false`

..args `array` or `arguments`

List of durations

prepa-duree

Converts decimal or fractional durations into integer durations

Parameters

```
prepa-duree(..args: array arguments) -> array
```

..args `array` or `arguments`

List of 2 to 4 numbers (d, h, min, s) to (min, s)

soustraction-durees

Sets up the subtraction of two durations.

You need to change units: ([j], [h], [min], [s]) with units: ([d], [h], [min], [s]).

Each duration is an array in one of the following forms:

- `(days, hours, minutes, seconds);`
- `(hours, minutes, seconds);`
- `(minutes, seconds).`

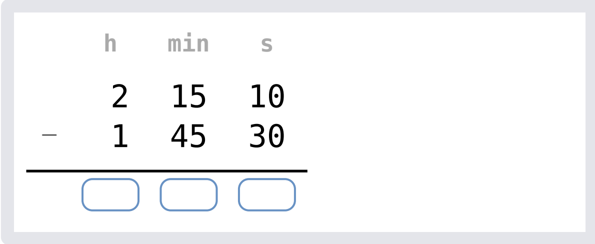
The first duration must be greater than or equal to the second.

Examples:

```
#soustraction-durees(
  (2, 4, 15, 10),
  (1, 6, 45, 30),
)
```

	j	h	min	s
		+24	+60	+60
	2	04	15	10
−	1	06	45	30
	+1	+1	+1	
	0	21	29	40

```
#soustraction-durees(
  (2, 15, 10),
  (1, 45, 30),
  hide-result: true,
)
```



Parameters

```
soustraction-durees(
  show-borrow: boolean,
  couleur: color,
  carry-size: relative length,
  size: relative length,
  convertion: boolean,
  hide-empty: boolean,
  hide-result: boolean,
  couleur-cadre: color,
  couleur-solution: color,
  type-mask: str,
  liste: array,
  solution: boolean,
  signe: boolean content,
  texte: relative length,
  show-units: boolean,
  units: array,
  units-size: relative length,
  couleur-unites: color,
  zero-pad: boolean,
  align: boolean,
  verticales: boolean,
  ..args: array arguments
) -> content
```

show-borrow `boolean`

Show borrows and carries.

Default: `true`

couleur `color`

Color of the borrows and carries.

Default: `red`

carry-size `relative length`

Size of the carries and the + and = symbols, max. 1em.

Default: `.7em`

size `relative length`

Width of the columns.

Default: `2.5em`

convertion `boolean`

Convert the minuend instead of displaying carries.

Default: `false`

hide-empty `boolean`

Hide columns containing only zeros.

Default: `true`

hide-result `boolean`

Hide the result.

Default: `false`

couleur-cadre `color`

Border color of hidden values.

Default: `blue.mix(gray)`

couleur-solution `color`

Color of values in the solution.

Default: `red`

type-mask `str`

Mask type: “rect” or “line”.

Default: `"rect"`

liste `array`

Indices of the cells to hide.

Default: `()`

solution `boolean`

Show the solution.

Default: `false`

signe `boolean` or `content`

Sign placed before the result.

Default: `false`

texte `relative length`

Text size.

Default: `1em`

show-units `boolean`

Show unit names.

Default: `true`

units `array`

Names of the four units.

Default: `([d], [h], [min], [s])`

units-size `relative length`

Unit size, max. 1em.

Default: `.75em`

couleur-unites `color`

Color of the units.

Default: `gray`

zero-pad `boolean`

Display hours, minutes, and seconds using two digits.

Default: `true`

align `boolean`

Align the digits of the units.

Default: `true`

verticales `boolean`

Vertical dotted lines to separate the columns.

Default: `false`

..args `array` or `arguments`

List of durations

Step-by-step calculations and highlighting

- detail()
- etapes-calcul()

detail

Function for writing calculation steps manually.

```
#detail("A=15/21=(5*r3)/
(7*r3)=5/7",arrondi:true,c:"circle")
```

$$A = \frac{15}{21} = \frac{5 \times 3}{7 \times 3} = \frac{5}{7} \approx 0.71$$

```
#detail("B=(15s2)/(21s18)=(5*r3 s2)/
(7*r3 s(9*2))=(5 ps2)/(21 bs2)
=5/21",c:"color",vertical:true,
arrondi: true)
```

$$\begin{aligned} B &= \frac{15\sqrt{2}}{21\sqrt{18}} \\ &= \frac{5 \times 3\sqrt{2}}{7 \times 3\sqrt{9 \times 2}} \\ &= \frac{5\sqrt{2}}{21\sqrt{2}} \\ &= \frac{5}{21} \\ &\approx 0.24 \end{aligned}$$

```
#detail("cal(A)=pi * r^2 = pi*5^2=25
pi",arrondi: true,digits: 4)
```

$$\mathcal{A} = \pi \times r^2 = \pi \times 5^2 = 25\pi \approx 25 \times 3.1416 \approx 78.54 \approx 78.54$$

Parameters

```
detail(
  expr-str: str,
  vertical: boolean,
  arrondi: boolean,
  digits: int,
  appro: int,
  todo: boolean,
  space: length,
  a: color,
  b: color,
  f: color,
  g: color,
  r: color,
  l: color,
  m: color,
  n: color,
  o: color,
  t: color,
  p: color,
  y: color,
  c: str
) -> content
```

expr-str `str`

The calculation expression as described above

vertical `boolean`

Vertical layout

Default: `false`

arrondi `boolean`

Append a rounded value at the end

Default: `false`

digits `int`

Number of decimal places for the rounded value

Default: `2`

appro `int`

Step from which "=" is replaced by "≈" when appropriate

Default: `0`

todo `boolean`

If `todo:true`, only the initial expression is displayed

Default: `false`

space `length`

Spacing before the next calculation

Default: `1fr`

a `color`

`a`: aqua; can be changed from Typst's default color if needed

Default: `aqua`

b `color`

`b`: blue; can be changed from Typst's default color if needed

Default: `blue`

f `color`

`f`: fuchsia; can be changed from Typst's default color if needed

Default: `fuchsia`

g `color`

`g`: green; can be changed from Typst's default color if needed

Default: `green`

r `color`

`r`: red; can be changed from Typst's default color if needed

Default: `red`

l `color`

`l`: gray; can be changed from Typst's default color if needed

Default: `gray`

m `color`

`m`: maroon; can be changed from Typst's default color if needed

Default: `maroon`

n `color`

`n`: rgb(8,111,189) (navy); can be changed if needed

Default: `rgb("#086fbd")`

o `color`

`o`: orange; can be changed from Typst's default color if needed

Default: `orange`

t `color`

`t`: rgb(0,128,128) (teal); can be changed if needed

Default: `rgb(0,128,128)`

p `color`

p: purple; can be changed from Typst's default color if needed

Default: `purple`

y `color`

y: yellow; can be changed from Typst's default color if needed

Default: `yellow`

c `str`

Display mode: "cancel", "circle", or "color"

Default: `"color"`

etapes-calcul

Show the step-by-step evaluation of `expr-str` (unless `todo:true`), with the option to perform independent operations of the same precedence simultaneously using `concomitant`, display the work vertically or horizontally, highlight the current step (`highlight`), and use fraction mode to keep computations exact by forcing calculations to continue with fractions.

```
#etapes-calcul("[ 6 + 5 * (23 + 8 /
2) ] * 8", fraction: false, vertical:
true,n:0)
```

$$\left[6+5\times\left(23+\frac{8}{2}\right)\right]\times 8$$
$$= [6+5\times(23+4)]\times 8$$
$$= [6+5\times 27]\times 8$$
$$= [6+135]\times 8$$
$$= 141\times 8$$
$$= 1128$$

```
#etapes-calcul("(200 - 45 * 2) / ((4 *
7 - 3 : 6) * 2)",concomitant: true)
```

$$\frac{200-45\times 2}{(4\times 7-3\div 6)\times 2}=\frac{200-90}{(28-0{,}5)\times 2}=\frac{110}{27{,}5\times 2}=\frac{110}{55}=2$$

```
#etapes-calcul("2 * pi * 3", fraction:
false, name: "B",n:3)
```

$$\mathrm{B}=2\times\pi\times 3=2\pi\times 3=6\pi\approx 18{,}85$$

```
#etapes-calcul("ln(exp(1)) + 3",
fraction: true, name: "E")
```

$$\mathrm{E}=\ln(\exp(1))+3=1+3=4$$

```
#etapes-calcul("sqrt(8)*sqrt(27)",
fraction: true)
```

$\sqrt{8} \times \sqrt{27} = 2\sqrt{2} \times \sqrt{27} = 2\sqrt{2} \times 3\sqrt{3} = 6\sqrt{6}$

Parameters

```
etapes-calcul(
  expr-str: str,
  mode: str,
  concomitant: boolean,
  vertical: boolean,
  digits: int,
  n: int,
  name: str content,
  todo: boolean,
  space: length,
  fraction: boolean,
  highlight: color boolean none,
  skip: array
) -> content
```

mode `str`

"fr" or "en" mode for the decimal separator

Default: `"fr"`

concomitant `boolean`

Set to true to perform independent operations of the same precedence simultaneously

Default: `false`

vertical `boolean`

Display the calculation vertically

Default: `false`

digits `int`

Number of decimal places for calculations involving π, etc.

Default: `2`

n `int`

Number of steps (counted from the end) at which "=" is replaced by "≈" when appropriate

Default: `0`

name `str` or `content`

Label to prepend to the expression

Default: `""`

todo `boolean`

Exercise mode: only the initial expression followed by “=” is displayed

Default: `false`

space `length`

Horizontal spacing between two `etapes-calcul` outputs

Default: `1fr`

fraction `boolean`

In fraction mode, computations remain exact, either as fractions or simplified forms such as 5π

Default: `false`

highlight `color` or `boolean` or `none`

Either a color used to highlight the current operation, `false` for bold, or `none` for no highlighting

Default: `rgb("#1e88e5")`

skip `array`

List of steps to skip if needed

Default: `()`